

M. Sc. Physics (Semester-II) 2026

Quantum Mechanics II – Tutorial 0

1. Calculate the commutation relation between the operators $[\hat{L}, \vec{x} \cdot \vec{p}]$. Here $\vec{L} = (L_x, L_y, L_z)$, $\vec{x} = (x, y, z)$, and $\vec{p} = (p_x, p_y, p_z)$ are operators and $L_i = \epsilon_{ijk} x_j p_k$.
2. Consider a vector $\vec{V}$ with components V_i ($i = 1, 2, 3$). When we rotate $\vec{V}$ to $\vec{V}'$, the three components become some other set of numbers, V'_i . The linear transformation, R , between the components is written as $V'_i = R_{ij} V_j$ (with Einstein summation convention). The norm of $\vec{V}$ is invariant under R i.e., $|\vec{V}|^2 = V_i V_i = |\vec{V}'|^2$. From here find condition satisfied by R and show that such matrices R form a group. Verify all group properties with multiplication as group operation.
3. If we associate a right-handed screw with a rotation, a positive ϕ rotation around the z -axis means that the screw is advancing in the positive z -direction. With this convention, write the form of rotation matrix $R_z(\phi)$, which rotates the coordinates (x, y, z) around z -axis. Similarly, write $R_x(\phi)$ and $R_y(\phi)$. R_x, R_y denote rotations around x, y axis, respectively.
4. Consider a rotation operation R around z -axis by an angle θ , characterized by a 3×3 orthogonal matrix R , we associate an operator $\hat{\mathcal{D}}(R_z(\theta))$ in the appropriate ket space such that

$$|\alpha\rangle_R = \hat{\mathcal{D}}(R_z(\theta))|\alpha\rangle$$

where $|\alpha\rangle_R$ and $|\alpha\rangle$ stand for the kets of the rotated and original system, respectively. Consider the case $j = 1$

- (a) What is the dimension of associated Hilbert space?
 - (b) What is the basis for this Hilbert space? (Hint $[J_z, J^2] = 0$)
 - (c) What are the components of $\hat{\mathcal{D}}(R_z(\theta))$ in this basis?
5. Find the matrix elements of $\hat{\mathcal{D}}(R_z(\theta))$ in the angular momentum basis $|j, m\rangle$, where j is integer.